A mere proposition is equivalent to its propositional truncation

The Univalent Foundations Program

hott-book:logic.tex:local-snapshot-20261012; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author lem environment at source lines 809-811 and its proof environment opens on line 812, the next line; the pairing is by adjacency and the retained proof range 812-816 is the whole environment.

Editorial note. The pinned book label-number dump carries no entry for this label, so no counter value is imposed and the counted lemma prints with the wrapper numbering; the original number is not claimed.

Editorial note. One label the proof cites is declared in external_references as a non-load-bearing navigation pointer: lem:equiv-iff-hprop in this chapter, which supplies the quasi-inverse criterion; it is counted by its own card of this lane.

Editorial note. The sentence introducing the lemma in the source calls the observation trivial; that framing sits outside the retained span and does not affect the proof range, which is retained line for line.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: a mere proposition is equivalent to its truncation (author label thm:prop-equiv-trunc).

Allowed standard prerequisites (editorial).

- The universal property of truncation. A map from a type into a mere proposition extends to a map from its truncation, and the extension sends a projection to the value of the original map.. It produces the map from the truncation of P back to P out of the identity map of P.
- Mere propositions and their path spaces. In a mere proposition any two elements are equal, and a map between mere propositions is an equivalence as soon as it has a section on one side.. It is what makes the two maps quasi-inverses and hence the comparison an equivalence.

Source reference. Original lemma 3.3.3: sources/hott-logic/pinned-original-source.tex, source lines 294–294. Book Lemma 3.3.3, the criterion that a map between mere propositions with a section is an equivalence, used to close the comparison. Navigation pointer only: it is counted by its own card of this lane.

Lemma 0.0.1. *If P is a mere proposition, then $P \simeq \|P\|$.*

Proof. Of course, we have $P \to \|P\|$ by definition. And since P is a mere proposition, the universal property of $\|P\|$ applied to $\mathsf{id}_P : P \to P$ yields $\|P\| \to P$. These functions are quasi-inverses by original lemma 3.3.3. □